

Branson Tay

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WORK EXPERIENCE

Jaguar TCS Racing

Aug. 2024 – Aug. 2025

Junior Strategy & Software Engineer (Placement)

Kidlington, United Kingdom

Stack: Python, MATLAB, ReactJS, Typescript, CSS, MongoDB, AWS, Azure Pipelines, Docker, VBA

- Create energy regeneration and friction brake sensitivity matlab tool allowing strategists to make informed calls.
- Analysed ATLAS race telemetry to optimise energy management and performance to tangibly reduce lap time.
- Pioneered a data processing python-pipeline converting noSQL-stored data to processed telemetry channels.
- Wrote custom library with ATLAS API to read and write raw data, defining data architecture for use in Python.
- Designed REST API for streaming of live race telemetry from WinTax using Visual Basic Scripts for race team.
- Collaborate in ML/AI pipeline planning alongside data scientists enabling AI projects and ML model training.
- Develop full-stack ReactJS data visualisation apps focused on UX for engineers to read time-critical race data.
- Deployed CI/CD pipeline for trackside executables ensuring software reliability via error handling and testing.
- Architected containerised applications using Docker with scope for CI pipelines and reliable deployment.
- Ensure battery temperature simulation correlation with track and simulator data for race temperature prediction.
- Assisted in race-weekend troubleshooting for web-based race-critical tools, ensuring data processing reliability.
- Update work items in Azure DevOps ensuring alignment with relevant stakeholders through discussion.
- Perform competitor systems radio analysis (energy management, pit-stops, strategy, battery temperature management, vehicle setup and tyre run-plans) communicating concise per-session summaries to strategists.

Republic of Singapore Navy

Nov. 2020 – Jul. 2022

Marine Systems Specialist Technician

Singapore, Singapore

- Trained in 4-stroke and 2-stroke engines, transmission, HVAC, water cooling, electrical and charged-air systems.
- Underwent 6-month intensive military training to operate as a technician independently and collaboratively.
- Led remote war-time firefighting simulation aboard ship engine operations room to assign appropriate fixes.
- Analysed live ship telemetry of core ship systems to handle damage control under high-pressure situations.
- Contributed to 50 pre- and post-mission debriefs, highlighting potential improvements for team efficiency.

Travelindr

Oct. 2019 – Mar. 2020

Co-founder & CEO

Singapore, Singapore

- Founded Travelindr, a group travel-itinerary generator, in a cross-functional team implementing Agile practices.
- Rapidly iterate on prototype through focus groups using A/B testing ensuring E2E user-integrated app design.
- Pitched to investors, achieved Minimal-Viable Product status, incubated in esteemed multi-national company.

EDUCATION

University of Leeds

Sep 2022. – Jul. 2026

BSc Computer Science with Year in Industry

Leeds, UK

- **Formula Student Performance & Simulation Sub-team Lead:** Lead development of simulation-driven projects in aerodynamics, tyres, powertrain, and vehicle dynamics, via simulation analysis for design choices.
- **Final Year Project:** Built a race strategy simulation platform to investigate the “competition vs. cooperation” metric in Multi-agent Reinforcement Learning investigating agent learning based on reward or training structure.

Ngee Ann Polytechnic

Jan. 2017 - Mar. 2020

Diploma in Mechanical Engineering

Singapore, Singapore

- Developed a gearbox meeting specific load, speed, and dimensional requirements for efficient material handling.
- Utilised AutoCAD CAD software to fabricate gearbox given specific load, speed, and dimensional requirements
- Recognized with the Good Progress Award and the School of Engineering Merit Award.

TECHNICAL PROJECTS

Formula Student Lap Time Simulator | Python, Vehicle Dynamics, Simulation Modelling

- Design modular steady-state lap time sim using point-mass and bicycle models for design sensitivity analysis.
- Combined physics-based and empirical models to distill complex racing dynamics into tuneable components.
- Created tyre model with Pacejka magic formula, ensuring model performance accurately correlates to real data.

Radiator Thermal Management Simulator | Matlab, MotecI2

- Constructed radiator size optimiser to correlate real-world and simulator thermal impact on the coolant system.

2D CFD Formula Student Radiator Optimisation Simulator | Rust, Numerical Methods

- Developed a 2D incompressible fluid solver analysing radiator angle effects on cooling, pressure drop and drag.
- Automated batch simulations and visualisations to optimise radiator design, leveraging parallel processing.

Formula Student ICE Data Acquisition System | C++, Git, Arduino

- Built a real-time GPS telemetry pipeline from Arduino Teensy to dashboard and wireless engineering displays.
- Honed proficiency in collaborative software engineering tools like Git and conducting code reviews.

Formula Student EV Battery Management System Emulator | C++, CAN Bus protocol

- Architected an expansion module using the CAN bus protocol for robust and efficient data transfer.
- Designed microprocessor framework to process battery module voltages to tangible data in C++.

ADDITIONAL EXPERIENCE

E-commerce Shopify Store Building | Facebook Ads Manager, Adobe Photoshop, Adobe Premiere Pro

- Drove user attention by utilising good design principles like obvious action buttons, and consistent elements.
 - Increased conversions by 10% by adapting “add-to-cart” button position based on cursor heatmaps.
- Constructed efficient sales funnel, extracting data insights, and targeting trends, fully utilising data analysis tools.
- Developed and marketed multiple e-commerce shop brands by creating catchy, attention-grabbing creatives.

Startup Partnership Intern at CoHive, Global Entrepreneurial Internship Scholar

- Pioneered Voucher System project integrating cross-department efforts within a strict one-month window.
 - Achieved a 10% surge in the 30,000 annual customer base nationally.
- Independently secured over 10 successful strategic partnerships through pitches with external senior executives.

TECHNICAL SKILLS, SOFT SKILLS & CERTIFICATIONS

Languages: Python, Matlab, HTML, CSS, Typescript, VBS, MongoQL, Rust, Java, C++, C, C#.

Frameworks and Libraries: NumPy, SciPy, Bootstrap, Matplotlib, ReactJS, NodeJS, MongoDB, MySQL, Flask.

Tools: Docker, Git, Claude Code, Codex, Azure DevOps, Azure Pipelines, Visual Studio, Microsoft Suite, ATLAS.

Soft Skills: Collaborative | Time-management | Curious | Adaptable | Multiperspectivity

Certifications & Awards

- Completed Harvard’s introductory course to computer science, CS50.
- Runner-up in TRIVE Global Entrepreneurial Internship Program Startup Challenge.

HOBBIES & PERSONAL INTERESTS

- Engineering: Avid viewer of motorsport and airplane engineering videos | Karting | Agentic Engineering.
- Stress-relieving: Table-tennis | Badminton | Cooking | Painting | Running.
- Enriching: Reading fiction & non-fiction books | Podcasts & videos on behavioural science.

References available upon request